

INTEGRATED EXPERIMENTAL PROCEDURES FOR THE STATIC LOAD TESTING OF STEEL- CONCRETE COMPOSITE HIGHWAY DECKS

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Abstract. The static acceptance testing of road and highway bridge decks traditionally relies on load tests aimed at measuring global deflections. While essential, this information alone is insufficient to fully describe the structure's behaviour—particularly for composite steel-concrete decks, where the global response must be interpreted alongside the local deformations of the main girders and the behaviour of the bearings. This paper proposes an integrated experimental procedure for the acceptance testing of a highway overpass, combining multidimensional static load tests with dynamic characterization under service conditions. The testing campaign involved the coordinated use of inclinometers, strain gauges, displacement transducers, and accelerometers, enabling the reconstruction of the deck's deformed shape, the assessment of local stress levels, the monitoring of bearings, and the identification of key modal parameters via Operational Modal Analysis and Enhanced Frequency Domain Decomposition. The results demonstrate a response consistent with the applied load scenarios, negligible residual deformations, and dynamic parameters suitable for establishing an experimental baseline for future monitoring. This integrated approach strengthens the acceptance assessment by reducing interpretive uncertainty compared to procedures based solely on deflection measurements.

Keywords: Static load testing, Composite steel-concrete decks, Load tests, Operational Modal Analysis, Enhanced Frequency Domain Decomposition, Structural monitoring.

1 Introduction

The static acceptance testing of road structures marks the final stage of the process leading from design to the structure's entry into service. At this stage, it is not enough merely to verify the physical construction of the work; it is also necessary to confirm—through a technical assessment based on experimental evidence—that the structural response aligns with design assumptions and expected performance levels.

For bridge decks, load testing has traditionally been the primary tool supporting the acceptance decision. In practice, however, assessments often focus on mid-span deflections, treating them as a summary indicator of overall behavior. While deflection is a significant parameter, it does not provide a complete picture of the deck's response—

particularly when the aim is to verify local behaviour and the actual functioning of bearing devices.

The Technical Standards for Construction and the associated explanatory circular emphasize the need to conduct tests that monitor the evolution of deformations. For major structures such as bridges and viaducts, they also require supplementing the acceptance test with additional investigations, including dynamic characterization tests [1, 2]. This approach calls for procedures that combine global, local, and dynamic measurements, moving beyond assessments based solely on vertical displacement.

International literature on bridge monitoring and dynamic identification highlights how vibration-based techniques can provide information complementary to static tests, particularly for establishing baseline conditions and monitoring performance changes over time [3, 4, 8, 9]. Recent reviews of Structural Health Monitoring approaches also highlight the benefits of integrating displacement, strain, and acceleration measurements, as each class of sensor captures a different aspect of the structural response [9, 10].

The specific contribution of this study lies in applying this integrated approach to the static load testing of a composite steel-concrete highway overpass. The aim is not merely to present the results of a single testing campaign but to demonstrate the methodological value of a coordinated use of instrumented static tests and operational dynamic tests. The proposed procedure transforms the load test from a point-in-time check of a single parameter into a multidimensional, more robust, and thoroughly documented experimental process.

2 Case study: composite highway deck

The case study concerns an overpass on the A20 Messina-Palermo motorway, featuring a composite steel-concrete deck. The structural scheme consists of a continuous beam supported at three points, comprising two identical spans of 40 m each, for a total length of 80 m.

The cross-section consists of two steel I-girders spaced 5.00 m apart, with a constant depth of 2.00 m. The girders are connected by truss cross-bracing and lateral bracing systems, while the reinforced concrete deck slab—cast in situ over precast panels—acts compositely with the steel girders via Nelson-type shear studs. The total width of the roadway deck is approximately 10.00 m, including the carriageway, shoulders, edge curbs, and safety barriers.

Elastomeric bearings are used at the abutments and the central pier. Given this configuration, it was deemed appropriate to supplement the measurement of the global deformed shape with direct monitoring of displacements at the bearing locations, in order to assess the compatibility between the spans' structural response and the local behavior of the supports..

Table 1. Main geometric and structural characteristics of the case study.

Parameter	Value / description
Type	Highway overpass with composite steel–concrete deck

Structural scheme	Continuous beam on three supports
Spans	Two spans of 40 m
Total length	80 m
Main girders	Two steel I-girders, 2.00 m depth, 5.00 m spacing
Slab	Cast-in-place reinforced concrete slab, made composite through steel shear connector
Bearings	Elastomeric devices on abutments and pier

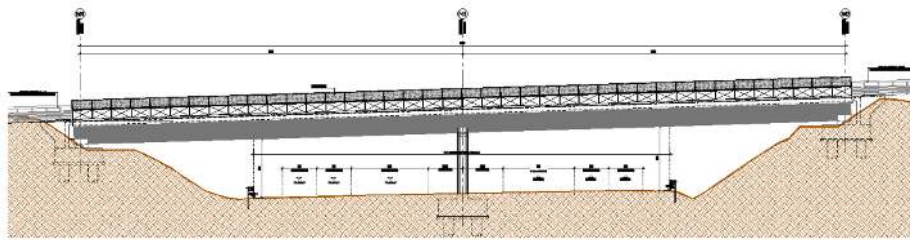


Fig. 1 General overview of the bridge deck.

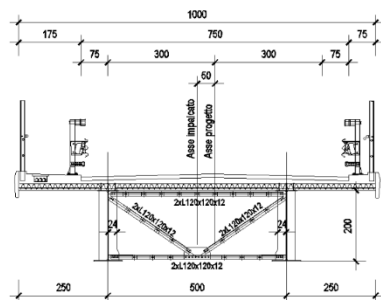


Fig. 2 Typical cross-section of the overpass

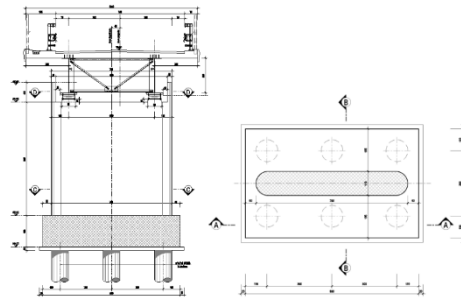


Fig. 3 Foundation section and plan of the pier

3 Integrated experimental procedure

The experimental procedure was designed to acquire—in a single, coordinated campaign—data representing the global behaviour, local response, and dynamic characteristics of the deck. The adopted strategy relies on three levels of observation: measurement of global deformation using inclinometers, direct measurement of local strains using strain gauges, and dynamic identification via accelerometers installed on the deck.

3.1 Static load tests

Static load tests were conducted prior to reopening the structure to traffic, following the standard procedure for the static acceptance testing of structural works [1, 2], using four-axle trucks with a gross weight of approximately 42 tonnes each, verified beforehand by weighing. Two main loading configurations were considered, defined to induce representative and mutually complementary static conditions.

In the first scenario, six trucks were positioned on a single span to maximize the positive bending moment—and thus the flexural response—of the loaded span. In the second scenario, eight trucks were distributed across the two spans in a 4+4 arrangement to increase the negative bending moment at the section above the central pier.

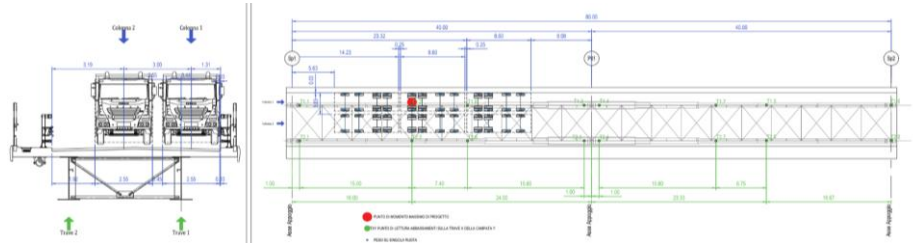


Fig. 4 Loading configuration: Scenario 1 — 6 trucks on a single span.

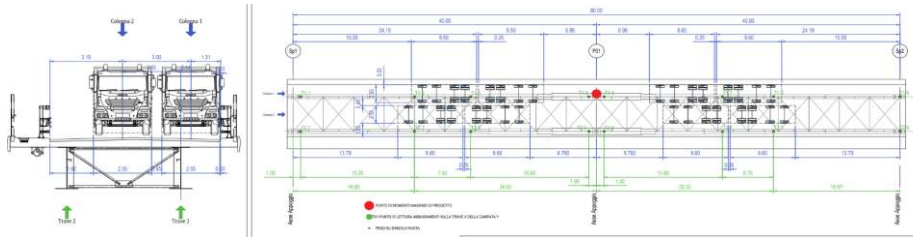


Fig. 5 Loading configuration: Scenario 2 — 8 trucks on two spans.

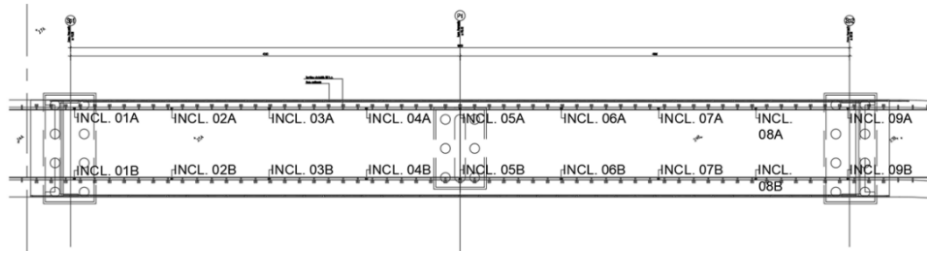
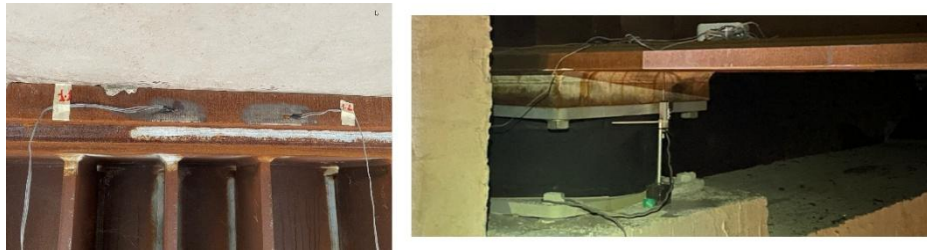
3.2 Static measurement system

The static measurement system was configured to go beyond an assessment limited solely to mid-span deflection. To this end, three types of sensors were employed, each associated with specific structural data.

Electronic inclinometers, installed symmetrically on both sides of the deck, enabled the measurement of elastic line rotations and the reconstruction of the deformed shapes of the two spans. Electrical strain gauges were applied to the top and bottom flanges of the main girders at the central support section to measure local strains in the extreme fibers and derive the corresponding stress levels. Finally, displacement transducers installed at the supports allowed for the recording of settlements and potential differential rotations of the elastomeric bearings.

Table 2. Summary of the instrumentation used for static tests.

System	Number / arrangement	Measured quantity	Function in the load test
Electronic inclinometers	18 sensors, distributed along both sides of the deck	Rotations	Reconstruction of the global deflected shape
Electrical strain gauges	8 sensors on the top and bottom flanges	Local strains	Control of extreme fibres and estimation of stress levels
Displacement transducers	3 sensors per bearing, arranged at 120 degrees	Local displacements	Monitoring of bearings and residual movements

**Fig. 6.** Inclinometers: plan view of the sensor layout.**Fig. 7.** Installation of strain gauges and transducers on the bearings.

3.3 Dynamic tests under ambient excitation

Dynamic characterization was performed under ambient excitation conditions—without the use of artificial forcing—following the Operational Modal Analysis approach [4]. This choice aligns with the need to describe the deck's global behaviour under near-service conditions and to establish an experimental baseline for comparison with future measurements.

The dynamic testing campaign utilized 15 PCB 393-A03 accelerometers connected to an NI cDAQ-9172 unit equipped with 24-bit USB-9234 modules. The final configuration comprised 12 vertical accelerometers distributed at the quarter-span and mid-

span points of the two spans, along with 2 transverse accelerometers and 1 longitudinal accelerometer at mid-span. Positioning the sensors along the vertical axes of the main girders prioritized the identification of the deck's global modes while minimizing interference from potential local vibrations of the lateral cantilevers.

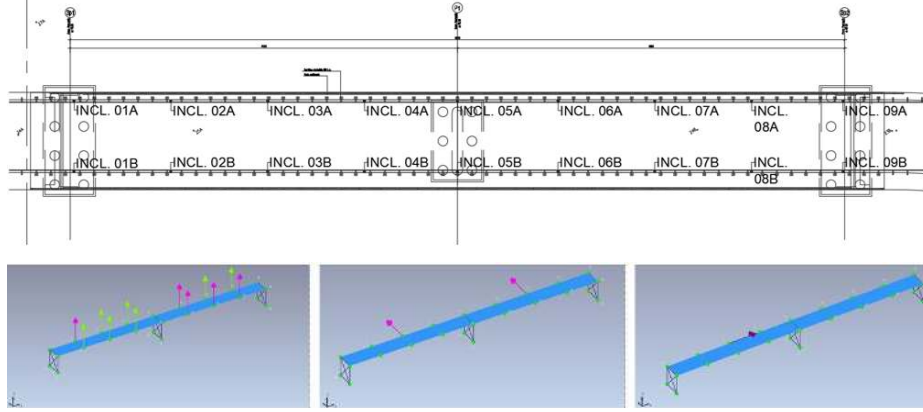


Fig. 8. Accelerometer placement layout and experimental model for dynamic analysis.

4 Processing of experimental data

4.1 Reconstruction of the deformed shape

Inclinometric measurements provide the discrete field of rotations along the axis of the deck. Assuming small rotations, the relationship between rotation and vertical displacement can be expressed by Eq. (1):

$$\theta(x) \simeq dv(x) / dx \quad (1)$$

The deformed shape is thus obtained by numerically integrating the rotation field, imposing the appropriate boundary conditions at the supports in accordance with Eq. (2):

$$v(x) = \int \theta(x) dx + C \quad (2)$$

In this way, the discrete measurement of rotation is transformed into a continuous description of the deformed configuration, useful for comparison with computational model predictions and for assessing the consistency of the global response..

4.2 Local deformations and stresses

The strains measured by the strain gauges were used to evaluate the local response of the top and bottom fibres of the main girders at the central support section. Assuming linear elastic behaviour for the steel, the stresses were estimated using Hooke's law, given in Eq. (3):

$$\sigma = E \varepsilon \quad (3)$$

where E represents the elastic modulus of the steel and ε the measured strain. This analysis makes it possible to verify not only the sign and distribution of the strains in relation to the loading scheme but also the order of magnitude of the stresses actually reached during the test.

4.3 OMA/EFDD modal identification

The accelerometer signals were processed using Operational Modal Analysis techniques [4]. In particular, the Enhanced Frequency Domain Decomposition algorithm was used, derived from the Frequency Domain Decomposition and based on the singular value decomposition of the spectral density matrix of the responses [6]. The estimation of the spectral densities was carried out using procedures based on the average of successive periodograms, according to the commonly adopted approach for the reduction of variance in spectral estimation [7].

In synthetic form, for each discrete frequency, the decomposition can be expressed as in Eq. (4):

$$S_{\gamma\gamma}(\omega_k) = U(\omega_k) \Sigma(\omega_k) U^*(\omega_k) \quad (4)$$

where $S_{\gamma\gamma}(\omega_k)$ is the power spectral density matrix of the responses, evaluated at the discrete frequency ω_k ; $U(\omega_k)$ is the unitary matrix whose columns are the singular vectors, taken as estimates of the mode shapes; $\Sigma(\omega_k)$ is the diagonal matrix containing the singular values ordered in descending order; and $U^*(\omega_k)$ represents the conjugate transpose of the matrix $U(\omega_k)$.

5 Experimental results

5.1 Global static response

The deformed shapes reconstructed from inclinometer measurements revealed a response consistent with the applied load configurations. In the first scenario, the greatest deflection is concentrated in the loaded span, while the adjacent span exhibits a deformed shape consistent with the continuity of the deck. In the second scenario, the load distribution across the two spans produces a more symmetrical response and subjects the central support zone to significant stress.

A comparative analysis of the two sides of the deck made it possible to assess the transverse regularity of the response. The differences observed between the Messina side and the Palermo side are consistent with the eccentricities of the load configurations and do not indicate any global anomalies in static behaviour. Furthermore, a joint examination of the four deformed shapes shown in Fig. 7 allows for the verification of response continuity between the two spans and the correct reversal of curvature in the zones near the central pier.

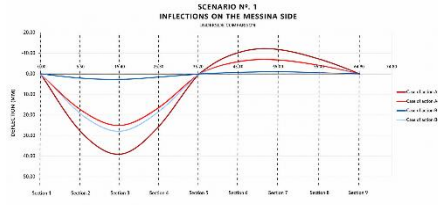


Fig. 9 Deflected configuration of the deck, Scenario 1 – Girder on ME side

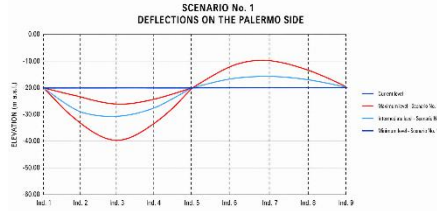


Fig. 10 Deflected configuration of the deck, Scenario 1 – Girder on PA side

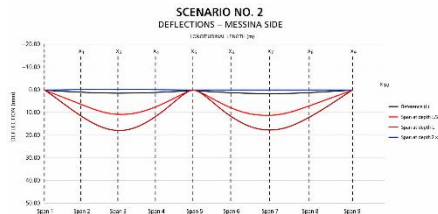


Fig. 11 Deflected configuration of the deck, Scenario 2 – Girder on ME side

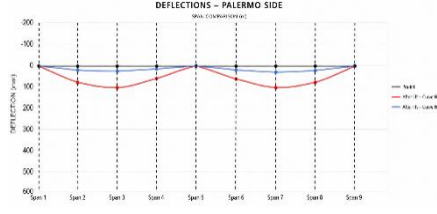


Fig. 12 Deflected configuration of the deck, Scenario 2 – Girder on PA side

5.2 Local deformations, stresses, and supports

Strain gauge measurements at the central pier section yielded trends consistent with the expected flexural behaviour. In particular, the strain distribution between the top and bottom flanges confirmed the sign reversal associated with negative bending moments in the support zone. Converting strains into stresses revealed values within the elastic range, confirming the absence of experimental evidence of non-linear behaviour during the testing phases.

Monitoring the bearings completed the assessment of the static response. Displacement transducer readings were low, and measurements taken after unloading showed negligible residual values compared to those at full load. The average maximum value recorded at full load was 0.42 mm, while the maximum residual value after final unloading was in the order of 0.05 mm. These findings make it possible to distinguish the deformation of the spans themselves from the local contribution of the bearing devices.

Table 3. Average support settlements recorded by displacement transducers.

Scenario	Phase	Bearing	Average reading [mm]
1	Full load	PA	0,42
1	Full load	ME	-0,32
1	Final unloading	PA	-0,02
1	Final unloading	ME	-0,04
2	Full load	PA	0,35
2	Full load	ME	-0,02
2	Final unloading	PA	0,01

5.3 Dynamic response

The OMA/EFDD analysis [4, 6] enabled the identification of the deck's principal global modes in the vertical, transverse, and longitudinal directions. Data processing was performed using Artemis Modal 7.2 by constructing a 24-node experimental model, comprising 12 nodes corresponding to measurement points and additional nodes representing external constraints. This representation allowed the mode shapes to be projected onto the observed degrees of freedom and the main modal families to be distinguished.

The identified vertical modes include flexural and torsional shapes, with frequencies ranging from 2.488 Hz to 7.614 Hz and damping ratios between 0.480% and 1.249%. Transverse and longitudinal modes—falling within a similar frequency range for the fundamental modes—complete the global dynamic description. The highest damping value was observed for the second longitudinal mode (8.750%), indicating greater dissipation associated with global motion components along the bridge axis.

Table 4. Main modal parameters identified using OMA/EFDD.

Direction	f [Hz]	zeta [%]	Mode description
Vertical	2,488	1,249	Out-of-phase bending
Vertical	3,643	1,011	In-phase bending
Vertical	4,002	0,850	Out-of-phase torsion
Vertical	4,487	0,970	In-phase torsion
Vertical	7,614	0,480	Double bending with mid-span inflection
Transverse	2,486	1,324	Out-of-phase transverse translation
Transverse	2,735	2,313	In-phase transverse translation
Transverse	3,633	1,061	In-phase transverse translation
Longitudinal	2,484	1,367	Longitudinal translation
Longitudinal	3,694	8,750	Longitudinal translation

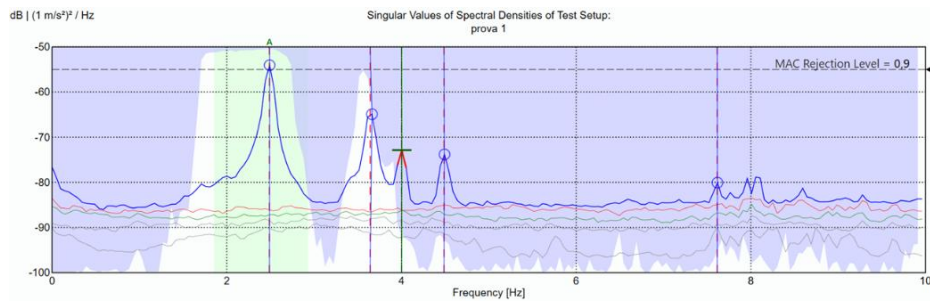


Fig. 13 Singular values of the PSD in the vertical direction.

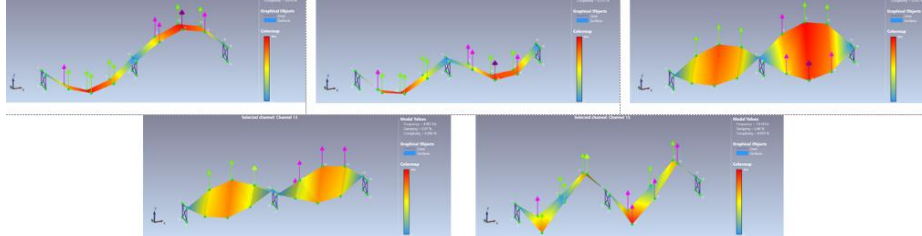


Fig. 14 Modal shapes obtained from the analysis of signals in the vertical direction.

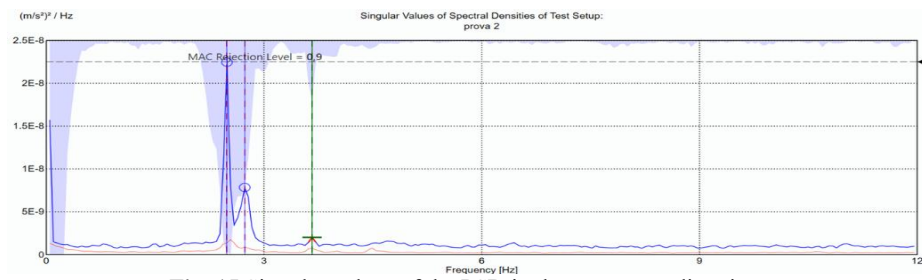


Fig. 15 Singular values of the PSD in the transverse direction.

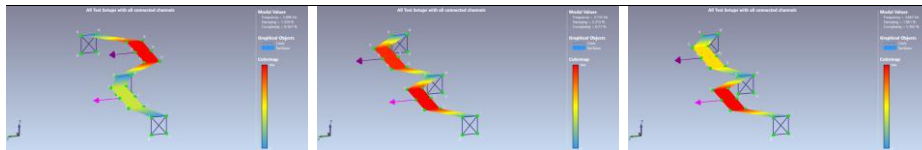


Fig. 16 Modal shapes obtained from the analysis of signals in the transverse direction.

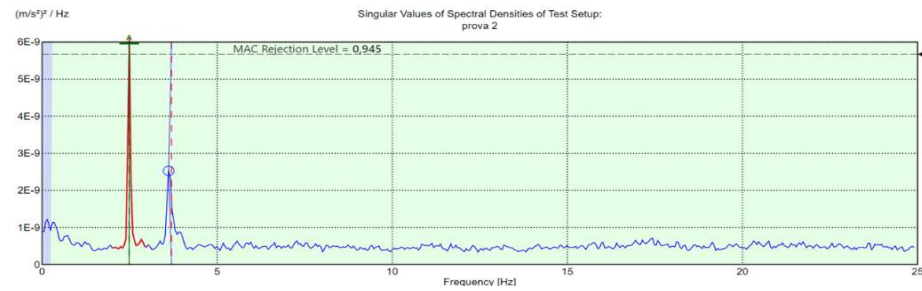


Fig. 17 Singular values of the PSD in the longitudinal direction.



Fig. 18 Modal shapes obtained from the analysis of signals in the longitudinal direction.

Table 5. Quantitative summary of the main experimental indicators.

Indicator	Observed value	Interpretation
Maximum average displacement of supports at full load	0,42 mm	Limited constraint response
Maximum residual displacement at final unloading	0,05 mm	Negligible residues
Identified vertical frequency range	2,488-7,614 Hz	Presence of global flexural and torsional modes
Vertical damping range	0,480-1,249%	Values consistent with global elastic response
Identified maximum damping	8,750%	Associated with the second longitudinal mode
Maximum average displacement of supports at full load	0,42 mm	Limited constraint response

6 Considerations regarding the integration of tests

Experimental evidence highlights that integrating static and dynamic tests yields a more comprehensive interpretive framework than relying on a single type of measurement alone. While static testing provides a direct verification of the structure's response to controlled loads, measuring deflections alone is not always sufficient to characterize the local behaviour of the most highly stressed sections and the support devices.

In this context, inclinometers allow for the reconstruction of the deck's global deformed shape, strain gauges enable the assessment of local deformations and stress levels in the main girders, and transducers installed on the bearings provide insights into support behaviour. Dynamic characterization completes the picture by identifying frequencies, mode shapes, and damping ratios—data that also serve as an experimental baseline for future monitoring [3, 4].

Unlike approaches found in the literature—where modal analysis is often treated as a standalone monitoring or model-updating activity [8, 9]—this study proposes integrating it directly into the acceptance testing procedure. In the case at hand, the consistency among static response, local deformations, bearing behaviour, and modal parameters strengthened the assessment of the structure's performance, thereby reducing interpretive uncertainty and transforming the acceptance test into a multidimensional experimental evaluation.

7 Conclusions

This study presented an integrated experimental procedure for the load testing of a composite steel-concrete highway bridge deck, based on the coordinated use of instrumented static tests and dynamic tests under ambient excitation.

The static tests—utilizing inclinometers, strain gauges, and displacement transducers—enabled a simultaneous assessment of the deck's global deformed shape, local

strains in the main girders, and the behaviour of the bearings. The results demonstrated a response consistent with the applied loading schemes, showing negligible residual displacements and stress levels within the elastic range.

The dynamic tests, analysed using OMA/EFDD techniques [4, 6], allowed for the identification of key modal parameters, providing a comprehensive description of the dynamic response and an experimental baseline for future monitoring [3, 8]. Integrating static and dynamic testing thus enhanced the reliability of the load test assessment, moving beyond evaluations based solely on deflection measurements.

The proposed approach is applicable to other road and highway bridge decks, particularly for structures requiring a simultaneous assessment of global response, local strains, support behaviour, and dynamic characteristics.

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